

Materials Letters

Development of an Intuitive Visualization System for Measuring Melt Electrowritten (MEW) Jet Diameter along the Spinline

--Manuscript Draft--

Manuscript Number:	
Article Type:	Short Communication
Keywords:	Software; 3D printing; Electrospinning; Electrohydrodynamics; Jet diameter
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Abstract:	Melt electrowriting (MEW) is a young and promising additive manufacturing technology for producing high-precision scaffolds. In this study, we develop an intuitive visualization system enabling users to measure MEW jet diameters along the spinline without programming knowledge. The digital measurement adopts the same theory as our reported manual measurement. The maximum error between the digitally and manually measured data is smaller than 9%, suggesting that the open-source system developed here is reliable in measuring jet diameters.
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28.5.2023

Submission of Manuscript

Dear Editor,

Please find our submission to *Materials Letters* entitled “**Development of an Intuitive Visualization System for Measuring Melt Electrowritten (MEW) Jet Diameter along the Spinline**” for consideration as a research article. Within the biomedical materials research community, melt electrowriting (MEW) is being increasingly used to make scaffolds. This submitted publication describes how to automatically achieve MEW jet diameter along the spinline without any programming knowledge.

MEW is a young, high-resolution 3D printing technology that has interest from the fabrication community, to make porous structures that are used for cell/tissue interactions. MEW uses an electrohydrodynamic effect to stabilize a molten polymer jet that thins out into a small diameter (micro-scale) fiber that is direct written onto a collector.

This is an essential submission to the field, which warrants consideration for a high-impact journal based on the state of the MEW field and the citation track record of previous papers. We published the work of manually measuring jet diameter along the spinline (Ashour, S., & Xu, H. (2022). *Polymers for Advanced Technologies*, 33(9), 3013-3016.), while it is time-consuming. The submitted manuscript uses the same processing steps as the published one, but we can achieve the data of jet diameter within 1-min processing. We have therefore emphasized this perspective, so that the submitted manuscript will become the seminal guide for current and future MEW researchers.

In assessing the quality and appeal of this manuscript for your journal, I highly recommend that you look, among other aspects, the Supporting Video provided.

Sincerely,

Huaizhong Xu

Assistant Professor

Kyoto Institute of Technology

1. An intuitive visualization system is developed for measuring MEW jet diameter.
2. The error between digitally and manually measured data is less than 9%.
3. A shooting system is updated to capture clear jet profiles.

Development of an Intuitive Visualization System for Measuring Melt Electrowritten (MEW) Jet Diameter along the Spinline

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Abstract

Melt electrowriting (MEW) is a young and promising additive manufacturing technology for producing high-precision scaffolds. In this study, we develop an intuitive visualization system enabling users to measure MEW jet diameters along the spinline without programming knowledge. The digital measurement adopts the same theory as our reported manual measurement. The maximum error between the digitally and manually measured data is smaller than 9%, suggesting that the open-source system developed here is reliable in measuring jet diameters.

Keywords

Software, 3D printing, Electrospinning, Electrohydrodynamics, Jet diameter

1. Introduction

Melt electrowriting (MEW) is an advanced additive manufacturing technology for printing high-resolution tissue engineering scaffolds [1]. With over one decade of development, numerous MEW-related devices have been established. For instance, researchers developed MEW devices that can print large-scale scaffolds [2], tubes [3], core-sheath filaments [4], etc. Meanwhile, the phenomena of jet lag [5], jet pulsing [6], fiber bridging [7], and layer shifting [8] were investigated, highlighting the viewpoint that jet control greatly influences print quality [9].

Currently, the jet lag angle [10] and the area of Taylor cone [11] are regarded as crucial indexes to judge the stability of printing throughout. Researchers can easily record jet lag angle and Taylor cone. In addition, jet diameter along the spinline is another critical parameter relating to the jet's kinematics and dynamics. We captured clear photos of the moving jet and manually measured the jet diameter along the spinline [12]. According to the mass conservation equation, we can estimate the jet velocities from the jet diameters. However, the current data processing method is time-consuming, and the results vary with each individual. In this context, we develop a web-based application to process the original image of the MEW jet and automatically output the jet diameter. Thus, it is convenient for MEW-related researchers to achieve the essential data without any programming knowledge.

2. Experimental

2.1 Updated Vision System

Figure 1 exhibits an updated vision system within a self-built MEW printer that was reported previously [13]. The primary change is the light source. We adopt an LED light (Brand: Yeezen) with changeable intensity. The light is fixed behind the print head. A 300× lens equipped camera (XW200, Shenzhen Hayear Electronics Co., Ltd., China) is fixed in front of the print head. Notably, the camera, nozzle, and light source should be placed in the same horizontal line.

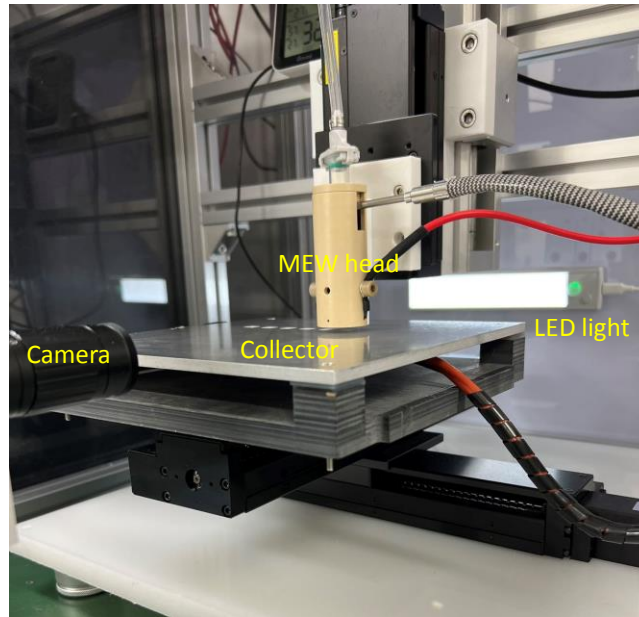


Figure 1. The updated vision system of the MEW printer.

2.2 Printing

Medical-grade polycaprolactone (PCL) with an inherent viscosity of 1.0 ~ 1.3 dl/g (Purasorb, PC12) is used for MEW. The fixed printing parameters are 22 G nozzle, 4 kV applied voltage, 3 mm nozzle-to-collector distance, 100 kPa air pressure, and 73 °C heating temperature. Under this condition, the critical translation speed (CTS) is 0.9 mm/s. The jet images were taken at collector speeds of 1×CTS, 2×CTS, and 3×CTS.

2.3 The Theory of Measuring Jet Diameter

The jet diameter is defined as the length of the line perpendicular to the jet's medial axis and ends by the jet edges. Specifically, the jet diameter is achieved after four processing steps, i.e., edge detection, skeleton (medial axis) extraction, fitting, and calculation.

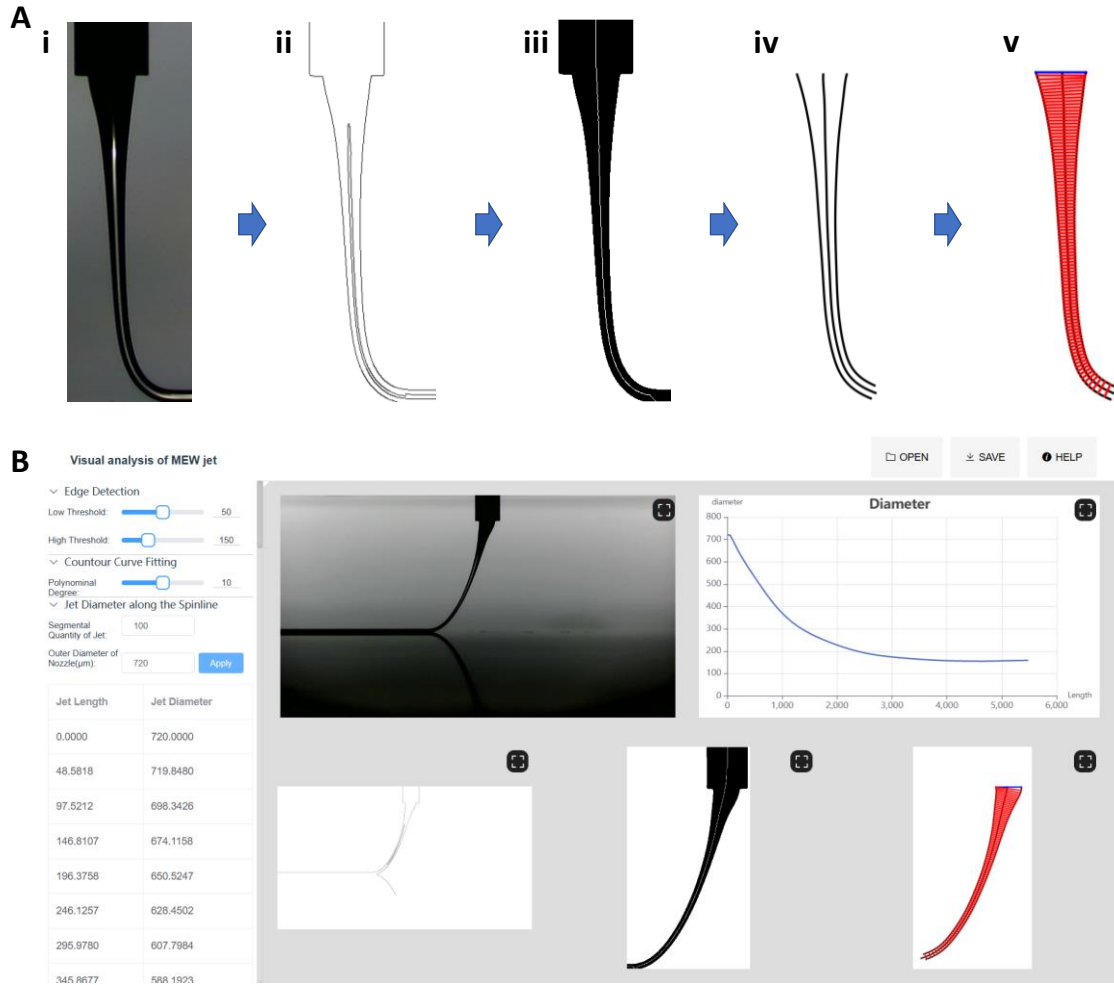


Figure 2. (A) A flow diagram of achieving jet diameters, including i) uploading an original image, ii) edge detection, iii) skeleton (medial axis) extraction, iv) fitting, and v) calculation. (B) A screenshot of the website (<http://124.220.224.64>).

- 1) Since edge detection is sensitive to image noise, we use Gaussian denoising to reduce the noise first and then use Canny edge detection to extract the edges after changing the original image to a binary image (**Figure 2A_ii**).
- 2) The skeleton extraction is realized by Zhang-Suen thinning algorithm. Since this algorithm uses an iteration to gradually delete the edge pixels, the jet in the binary image needs to fill in pixels (**Figure 2A_iii**).
- 3) The two processes mentioned above can achieve discrete data of the edges and the medial axis. Thus, it is necessary to smooth the discrete data to reduce the actual error after calculation. We

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4 apply polynomial equations to fit the curves, and the degree is set within 8 to 12 by trial and
5 error (**Figure 2A_iv**).

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8 4) The last step is calculating the jet diameter and the length of the medial axis (**Figure 2A_v**).

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10 The function of the normal of the medial axis (g_n) is

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$$g_n(x) = -\frac{x}{f'_{mid}(x_n)} + \frac{x_n}{f'_{mid}(x_n)} + f_{mid}(x_n) \quad (1),$$

14

15 where n is a positive integer, and $f_{mid}(x)$ is the function of the medial axis. When the line of the
16 g_n and the lines of $f_{left}(x)$ and the $f_{right}(x)$ (the functions of the left and the right jet profiles)
17 intersect, we can achieve the abscissas $x = x_{left_n}$ and $x = x_{right_n}$. Then the function of jet
18 diameter (d_n) can be expressed as
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$$d_n = \sqrt{(x_{right_n} - x_{left_n})^2 + (g_n(x_{right_n}) - g_n(x_{left_n}))^2} \quad (2).$$

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25 The length of the medial axis can be written as

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$$L(x) = \int_{x_0}^{x_n} \sqrt{1 + [f'_{mid}(x)]^2} dx \quad (3).$$

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30 At last, the actual jet dimension should be transformed according to the outer diameter of the
31 nozzle.
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35 36 2.4 System Interface Diagram

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38 **Figure 2B** displays the system interface diagram. After uploading a jet image, users can
39 automatically achieve a plot of jet diameter along the spinline. The interface also contains the
40 images after edge detection, skeleton extraction, fitting, and calculation. The source code for the
41 application used here can be accessed from an open-source repository (<https://github.com/whx-xyhf/visual-3D-printing>) and is hosted online for simplified access. (<http://124.220.224.64>). **Video**
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47 **S1** (Supporting Information) displays the operation of the software.
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3. Results and Discussion

Digitally measuring the jet diameter reported here uses the same procedure as manually measuring the diameter [12]. Thus, it is meaningful to compare their differences. CTS is a crucial index in MEW to guide the suitable print. The print speed is generally set between $1\times\text{CTS}$ and $3\times\text{CTS}$, so we compared the manually and digitally measured data for printing at $1\times\text{CTS}$, $2\times\text{CTS}$, and $3\times\text{CTS}$ (Figure 3).

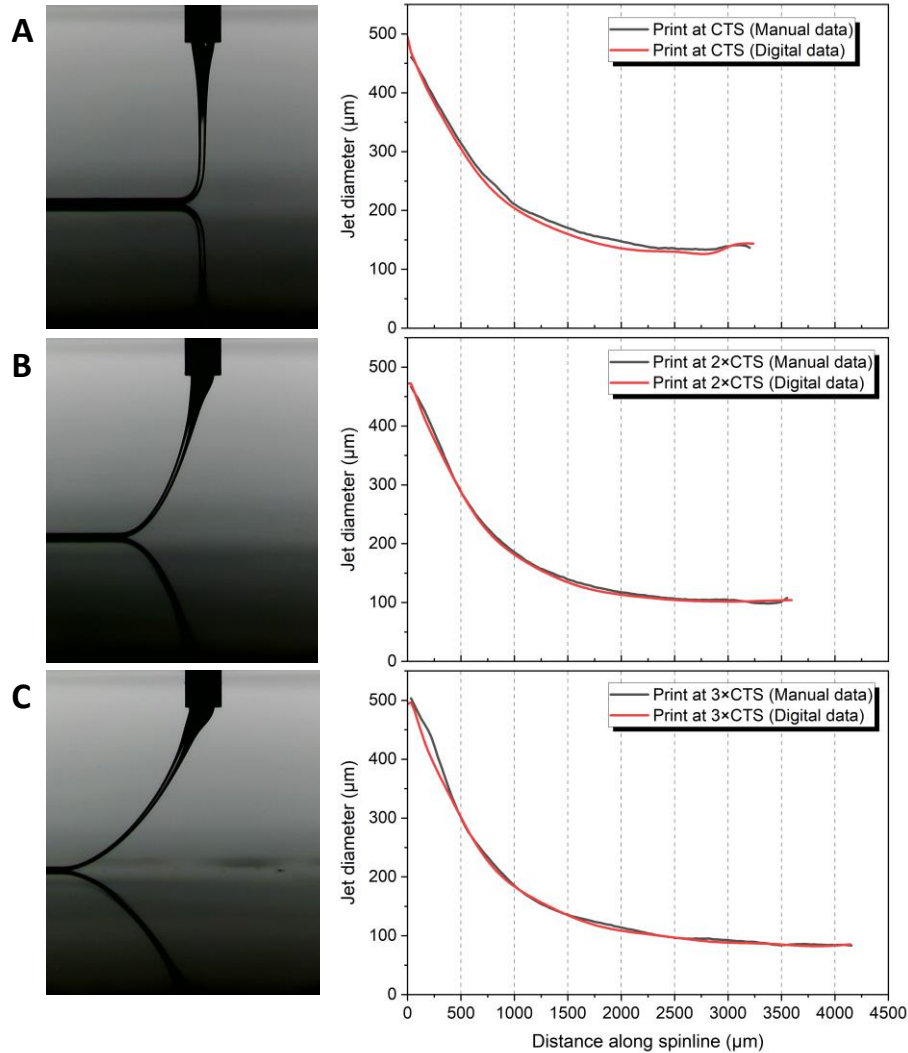


Figure 3. The manually and digitally measured jet diameter along the spinline. Jet is individually printed at (A) $1\times\text{CTS}$, (B) $2\times\text{CTS}$, and (C) $3\times\text{CTS}$.

The jet length is close to the nozzle-to-collector distance at $1\times\text{CTS}$, and the jet length increases with increasing the print speed because of the drawing process. Overall, the digital data match the manual data for different print speeds. To analyze their differences quantitatively, we must

compare the digitally and manually measured jet diameters at the same position. Instead of comparing the data of jet diameter processed from the fitting lines of the medial lines, comparing the data processed from the discrete point of the medial lines can reduce the error caused by fitting. Accordingly, the initial step is finding the same point along the spinline for the manually and digitally processed data. Herein we use the quadratic interpolation method. **Table 1** lists the errors between the manual and the digital data. We divide the length into three regions; that is, the Taylor-cone region ($L \leq 500$), the middle region ($500 < L \leq 2500$), and the tail region ($L > 2500$). For each jet segment, the maximum error is less than 9%, indicating that the developed software can replace the manual processing of measuring each jet diameter.

Table 1 Errors between the manually and the digitally processed data.

	Distance along the medial line (μm)	Mean absolute error (%)	Maximum error (%)	Minimum error (%)
1×CTS	$L \leq 500$	1.91	1.72	-2.82
	$500 < L \leq 2500$	5.39	-3.02	-8.40
	$L > 2500$	3.72	5.44	-5.93
2×CTS	$L \leq 500$	1.78	1.11	-3.32
	$500 < L \leq 2500$	2.79	-0.08	-4.73
	$L > 2500$	2.53	4.72	-3.62
3×CTS	$L \leq 500$	4.20	0.02	-8.38
	$500 < L \leq 2500$	2.13	1.74	-5.62
	$L > 2500$	2.46	2.75	-5.28

4. Conclusion

We successfully developed an accessible software enabling the measurement of jet diameter along its spinline. The digital processing is more efficient than the manual processing while can keep the data accurate. Meanwhile, we clarified the method of achieving clear jet images to enhance the quality of data extraction. It is a fundamental work of achieving basic jet data for further analysis of jet electrohydrodynamics.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (No.62277013, No.62177040), Zhejiang Provincial Science and Technology Program in China (No.2021C03137, No.QY19E050003), Public Welfare Plan Research Project of Zhejiang Provincial Science and Technology Department (No.LTGG23H260003).

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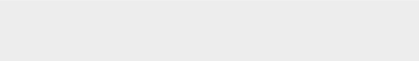
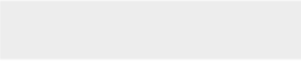
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Electronic Supplementary Material (online publication only)

Video S1.mp4



Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: